Research Brief (Gate G1) — Does the virtual tumor board need a board?

Vibhum Sharma (T32) • AI Tools for Research, Lab 7, Prompt 05 • 24 Oct 2026 • Labels: VERIFIED / INFERRED / UNKNOWN

Review topic and key papers	Lab 4 review: AI- and multi-agent-assisted virtual tumor boards (24 papers). Originals checked: Liu et al. 2026, EvoMDT, *npj Digit. Med.* (PDF, VERIFIED); Kim et al. 2025, multi-agent MDT, MLHC (PDF, VERIFIED); Park & Chae 2024, *J. Gastrointest. Cancer* (PDF, VERIFIED); Kim et al. 2024, MDAgents, NeurIPS (abstract and code VERIFIED; tables INFERRED); Wang et al. 2024, ACL (abstract VERIFIED).
Research gap	Review Sec. 7 (4th gap): multi-agent systems report the best results but have the thinnest independent validation. EvoMDT compares its DeepSeek-based board only with single-pass runs of *other* models (VERIFIED). Kim et al. 2025 tested roles and discussion on 31 private cases with an LLM judge and unequal call counts (VERIFIED). Untested: same backbone, **equal call budget**, public data, objective metric, small open model.
Research question	**RQ1:** with Qwen2.5-7B-Instruct fixed, is a 4-specialist tumor board (2 rounds + chair, 9 calls) more accurate than self-consistency with 9 samples on oncology exam questions? **RQ2:** does replacing specialists with generic agents change board accuracy? **RQ3:** does discussion change the vote, and how often does it turn correct members wrong vs. wrong members correct?
Study pathway	B — controlled comparative study, with C-style ablations (roles removed; discussion removed) taken from the same runs.
Dataset / protocol	MedQA-USMLE 5-option test (MIT, VERIFIED), oncology subset n = 97 by a frozen keyword filter (VERIFIED, hashed), pooled with MedMCQA dev oncology subset (MIT, VERIFIED; n UNKNOWN). Zero-shot: no training split, prompts debugged only on train items. Risk: benchmark contamination in pre-training (equal across arms).
Baseline / comparators	Same model for all arms: greedy CoT (1 call); self-consistency, 15 samples voted at $k = 1$, 3, 5, 9, 15 (SC@9 = call-matched); generic-agent board (ablation). Own implementation (MIT); MedAgents, MDAgents and Kim et al. code are unlicensed, so used only as references.
Primary metric and control	Accuracy (higher is better; unparseable = wrong). Exact two-sided McNemar, $\alpha = 0.05$, plus $\geq$ 2-point difference; paired bootstrap 95% CI. Held constant: model, prompts' answer format, T = 0.7, top-p = 0.95, max tokens, items, seeds. **Power** (simulated, assumed 15–25% discordance): n = 97 detects $\approx$ 12–15 points; n $\approx$ 400 detects $\approx$ 6–8 points. Nulls are reported against this limit.
Computing and risks	Kaggle 2×T4 (free), vLLM. Minimum run $\approx$ 1–3 GPU-h; full grid $\approx$ 5–20 GPU-h (INFERRED). Risks: vLLM on T4 untested (fallback: HF transformers); 7B too slow (fallback: Qwen2.5-3B); MedMCQA unavailable (fallback: MedQA oncology only, underpowered); quota (drop full MedQA, then seeds).
First executable milestone	*Done:* oncology subset built from the real MedQA file (1,273 $\rightarrow$ 97 items, sha256 recorded); filter audited before any model output; pipeline unit-tested with a mock model (29 tests pass). *Next:* Kaggle smoke test on 5 MedQA train items, measuring per-item cost and parse failures.

Contribution boundary. Claimed: a call-matched, public-benchmark test of the specialist-board idea with an open 7B model, and spontaneous error-propagation rates. Not claimed: a new method, clinical validity, or anything about real tumor-board decisions. A null result is a valid outcome.